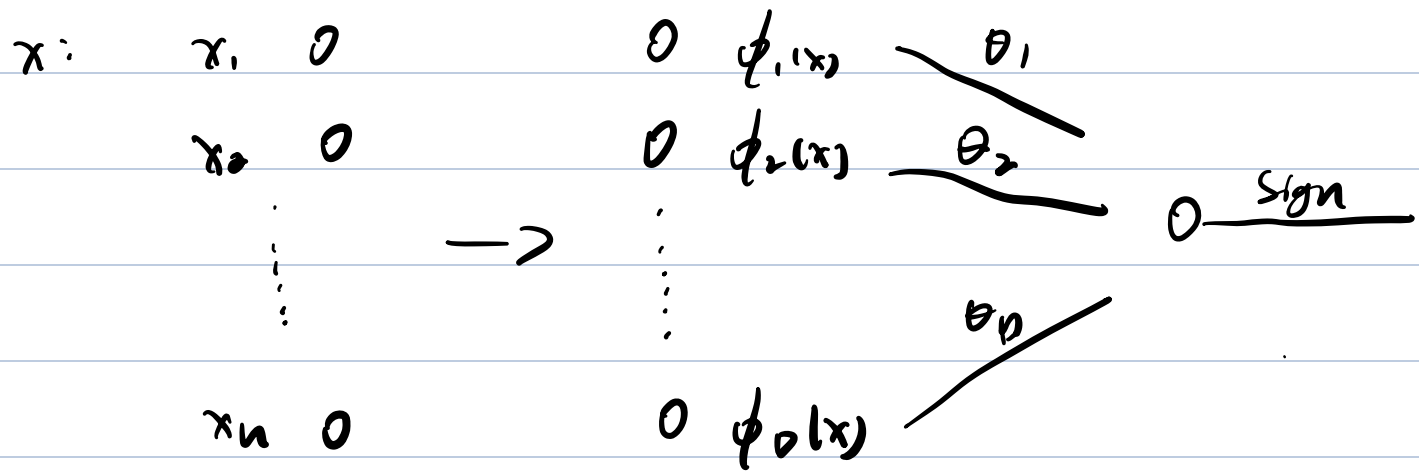
Unit 3

Feed-forward neural networks.

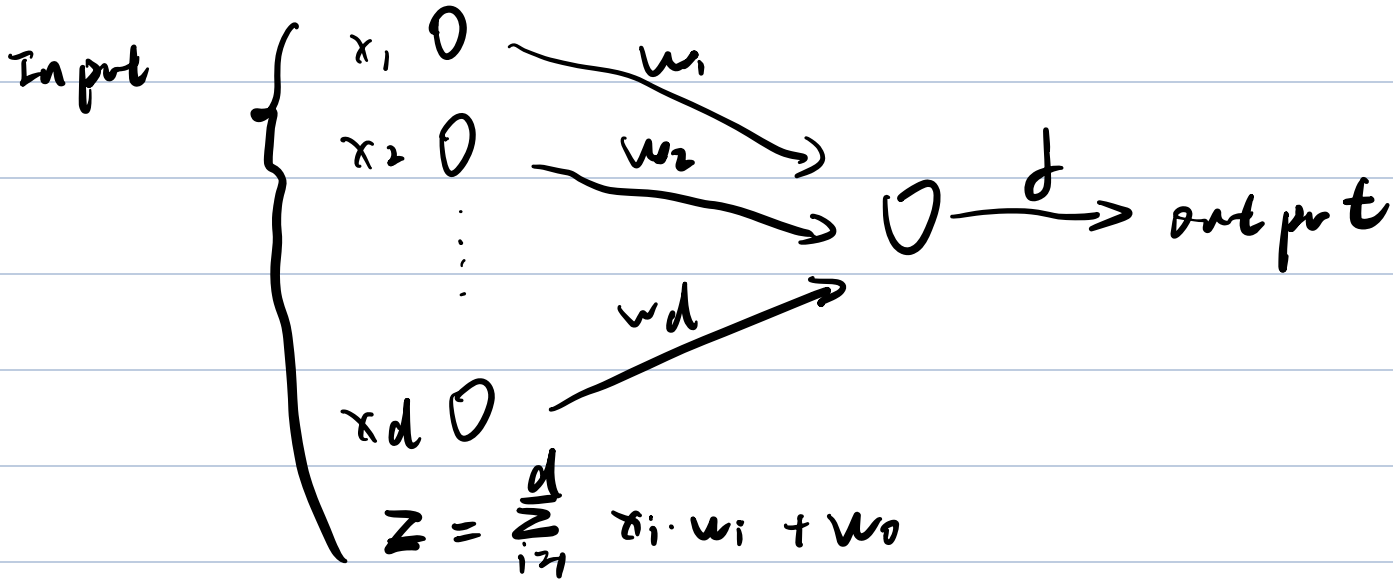
$$\hat{y} = \text{sign}(\theta \cdot \phi(x))$$



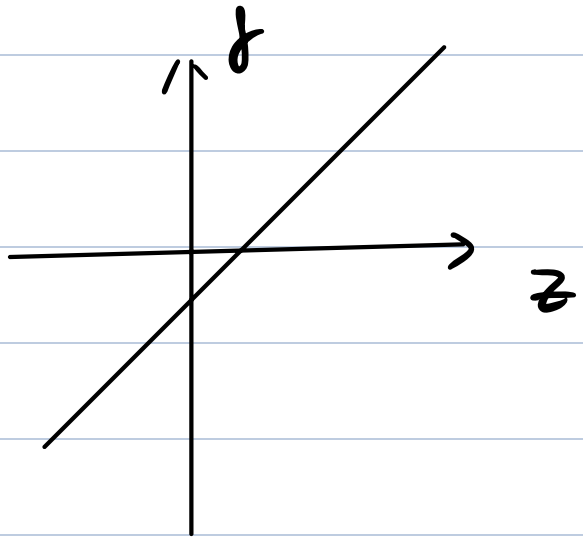
$$x \xrightarrow{\text{fixed}} \phi(x)$$

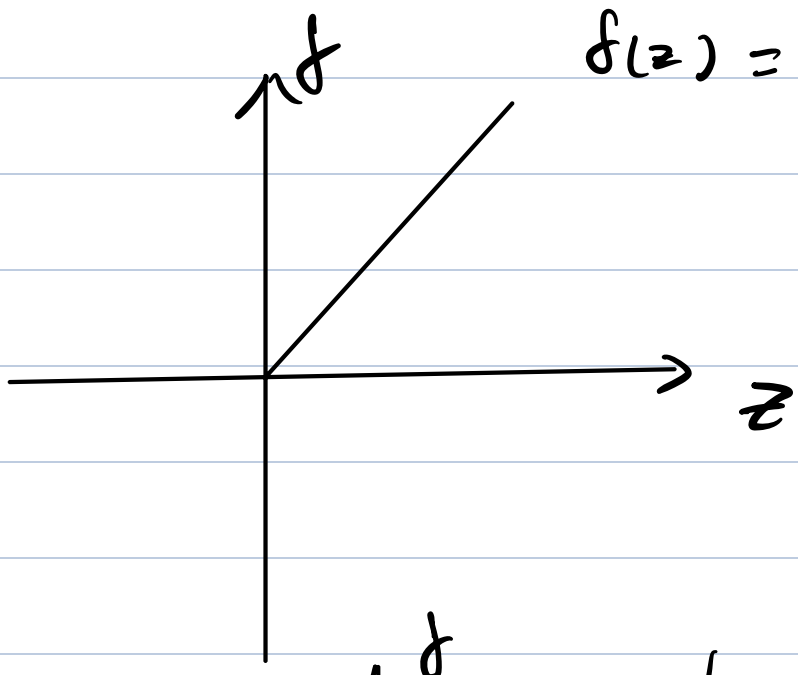
Units

w : weight

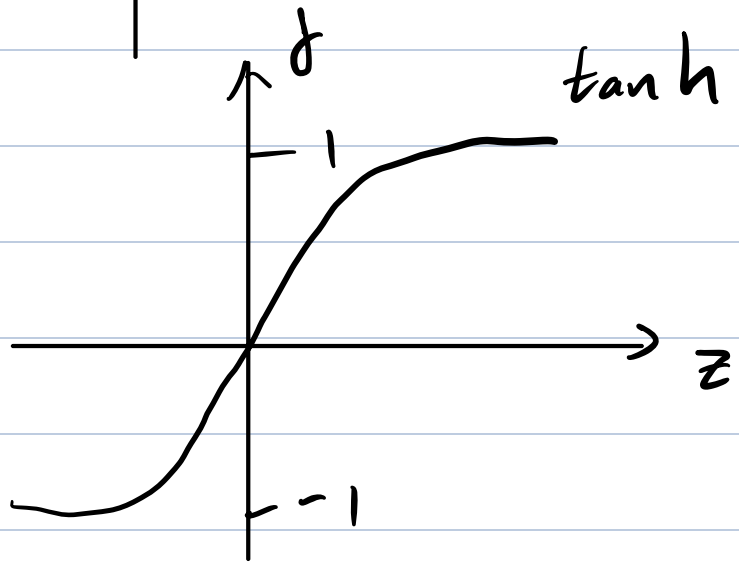


output: $f = f(z)$ (non-linear)

① 

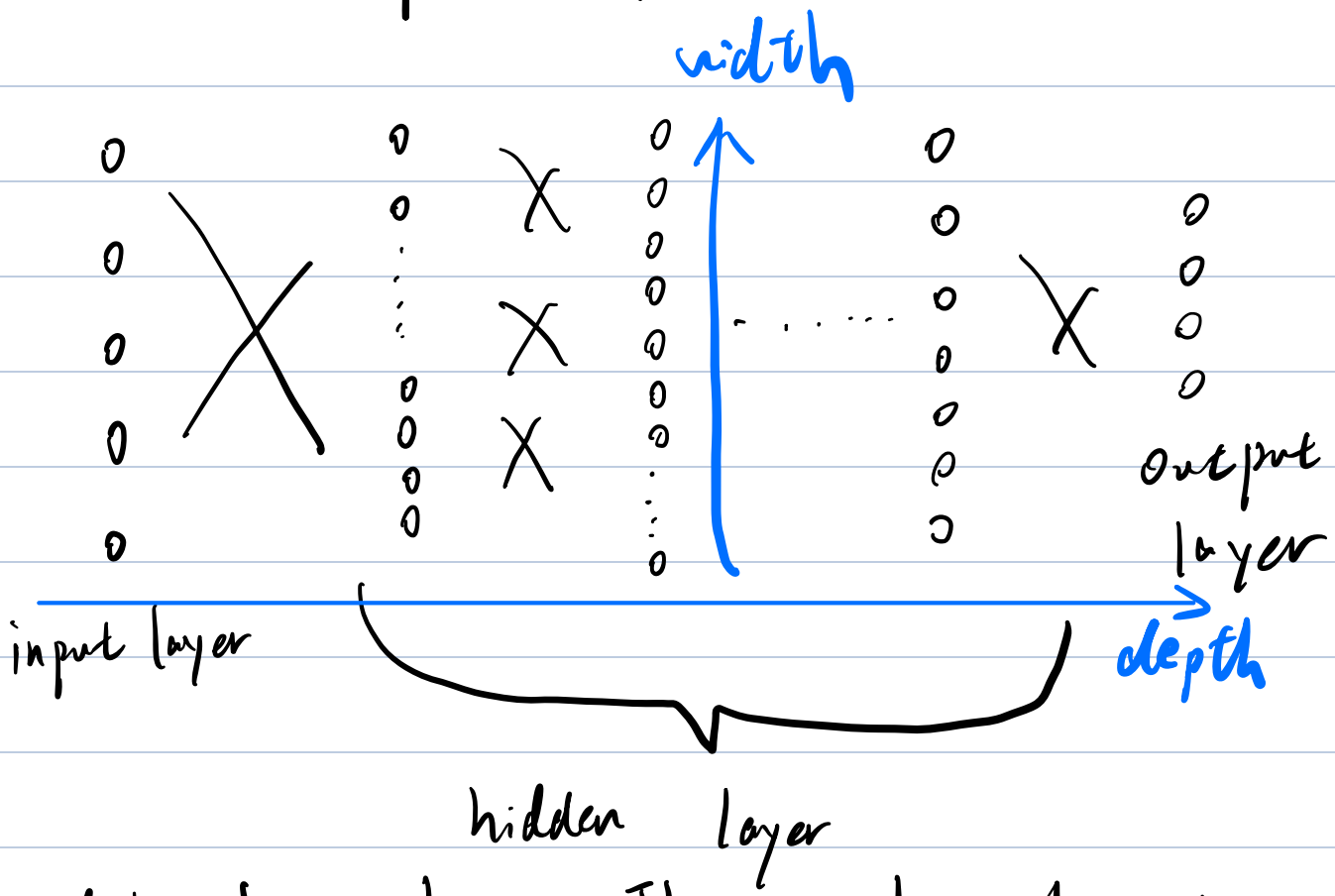
② 

$f(z) = \max(0, z)$

③ 

$\tanh(z)$

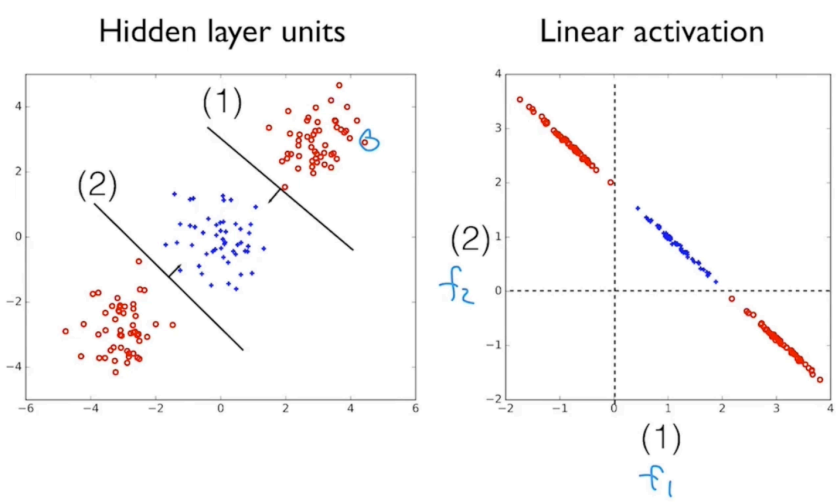
Deep Neural Networks



width of a layer: The number of units in that layer.

Depth of deep neural networks: The number of layers of transformations until we get to the answer.

Hidden layer representation



Relu : $f(x) = \max(0, x)$

Back - propagation Algorithm

1. Used in Feed - forward neural networks.

2. How it works: Evaluate the derivative of the Loss with respective of the parameters.

1. Compare $\text{Loss}(y, f(x; w))$
2. take derivative of the loss with respect to each of the parameters.

e.g. $\dfrac{d(\text{Loss}(y, f(x; w)))}{d w_{ij}}$

w_{ij} : the parameter of the j-th unit at the i-th layer

3. applying stochastic gradient descent.

$$w_{ij} \leftarrow w_{ij} - \eta \frac{d(\text{Loss}(y, f(x; w)))}{d w_{ij}}$$

Simple example

$$x \xrightarrow{w_1} \overset{z_1\ f_1}{\bigcirc} \xrightarrow{w_2} \overset{z_2\ f_2}{\bigcirc} \longrightarrow O$$

$x \in \mathbb{R}$

$z_1 = x \cdot w_1$

$f_1 = \tanh(x \cdot w_1)$

$\vdots$

$f_L = \tanh(f_{L-1} \cdot w_2)$

Aim: evaluate $\dfrac{d\,\text{Loss}}{d\,w_1} = \dfrac{d f_1}{d w_1} \cdot \dfrac{d\,\text{Loss}}{d f_1}$

$$\frac{d\,\text{Loss}}{d f_2} = \frac{d f_2}{d f_1} \cdot \frac{d\,\text{Loss}}{d f_2}$$

$\vdots$

$$\frac{d\,\text{Loss}(y; f_L)}{d f_L} = \ldots\ldots$$

with $\dfrac{d\,\text{Loss}}{d f_L}$, we can evaluate all of this